Tiny Yet Mighty: Quantized Deep Learning for Edge Plant Disease Detection

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***Abstract*—Plants are highly prone to diseases, which can lead to considerable losses in yield and quality, posing a significant threat to farmers' incomes. Recent solutions leveraging deep learning (DL) and computer vision (CV) have demonstrated strong potential for plant disease identification. However, such solutions overlook two pressing challenges: their applicability to uncontrolled real-world field environments where farmers operate, and their deployment feasibility. Most existing models have been trained on lab-controlled, benchmarked, homogeneous datasets. Under these ideal conditions, such models report near-perfect performance metrics; however, their applicability to real-world agricultural settings remains highly debatable. Moreover, such models are usually high-parameter, disk-heavy solutions, making them less suitable for deployment in resource-constrained environments. To address the above shortcomings, we propose a two-stage pipeline. For the first stage, the Xception model, with 1.06 million trainable parameters, was subjected to fine-tuning on the real-world, uncontrolled PlantDoc dataset via a two-stage training process that blended Bayesian hyperparameter optimization with adaptive fine-tuning, guiding performances that surpassed several state-of-the-art works. In the second stage, the trained Xception model undergoes INT8 quantization, reducing the disk size by 74.27%, with only $\sim 1\%$ drop in accuracy. Despite quantization, the model maintains substantial margins over existing methods, highlighting its potential for deployment in real-world, resource-constrained agricultural environments.**

***Index Terms*—Edge Intelligence, Model Quantization, Smart Farming, AIoT, Resource-Constrained Deployment**

I. Introduction

Agriculture has always been at the heart of human progress, shaping cultures by providing food through the ages and, even today, sustaining communities as a vital source of work and livelihood. With employability statistics suggesting the number is around a billion, this figure roughly accounts for about 28% of the employed people globally [1]. In terms of the most extensive net-cropped regions, the major players include India, China, & the USA [2]. Each facet during the cultivation of crops needs special attention for ensuring better yields year-round. Due to plant diseases, a staggering loss of $10 - 16\%$ is incurred worldwide, amounting to around 220 billion US dollars [3]. Additionally, the issue of food contamination caused by plant diseases has been a persistent problem that plant pathologists have been addressing and cannot ignore. Presently, around 83% of diseases infecting plants are caused by fungi, while viruses account for 9%, with bacteria accounting for 7% [4]. Thus, agrarian diseases are a practically imperative problem in crop production, affecting every field.

By long-standing practice, agricultural domain specialists use visible properties to inspect and diagnose plant diseases. This manual approach, however, is primarily prone to errors and inconsistent, specifically, in regions with resource limitations, along with inadequate expertise in agriculture [5]. However, such a process is time-consuming, subjective, and labor-intensive, limiting its implementation on a large scale. Additionally, environmental factors such as lighting and inspector expertise can significantly impact the accuracy of diagnosis [6]. Over the past few years, deep learning (DL)-based models because of their inherent feature learning capabilities apart from the generalization ability with regards to complex processes [7] have demonstrated their effectiveness as a solution for automating plant disease diagnosis, while ensuring both precision and speed in distinguishing between disease categories. Among the myriad of models under DL, convolutional neural networks (CNNs) have made a significant jump because of their notable know-how in image recognition, making them both well-suited and the default choice of the majority of the image-based solutions for plant diseases [8].Among CNN-based approaches, earlier works have demonstrated the effectiveness of adapting pre-trained models—such as Inception, MobileNet, and VGG—through transfer learning and fine-tuning [9].

Despite the significant achievements of DL-based models in plant disease detection, several key challenges remain. A major limitation lies in the reliance on datasets collected under controlled lab environments, which restricts their ability to generalize to real-world field conditions [10]. Although lightweight architectures such as MobileNet [11], DenseNet121 [12], and EfficientNetB0 [13] have been applied to this problem,

their performance often declines when tested on field-based agricultural data due to domain shifts and dataset biases. Another challenge concerns computational efficiency: many of these models remain parameter-heavy and require high-performance, hardware-aware devices, making them unsuitable for deployment on resource-constrained edge devices that are common in rural farming communities [14].

To address these challenges, our work takes a deployment-driven approach. While pre-trained CNNs have been widely explored for plant disease detection, their large size and heavy parameter requirements make real-world deployment difficult. In contrast, our fine-tuning strategy produces a lightweight model with significantly fewer parameters and smaller size, making it more practical for use in real farming contexts. Unlike earlier studies trained only on lab-controlled datasets, our model is trained on a field-collected, multi-crop dataset, ensuring stronger generalization to real conditions. We further introduce a quantized version that reduces disk size without compromising accuracy, enabling efficient deployment on edge devices. By combining efficiency, generalization, and edge readiness, our work aims to bridge the gap between advanced DL models and their practical application in agricultural machinery. The major contributions of our work include:

- A two-staged fine-tuning process is utilized, where Bayesian optimization systematically identifies compelling hyperparameter settings, followed by training refinement with early stopping, model checkpointing, and adaptive learning rate to facilitate stable and reliable convergence.
- Through fine-tuning of a deep CNN model, we propose a specialized approach over generic model building from scratch, achieving compact architectures at lower costs.
- Experiments have been conducted and validated on an uncontrolled field dataset, assuring model generalization to diverse real-world conditions.
- For real-world deployment, quantization has been applied to reduce the size of the model and computational needs, thereby facilitating its potential for usage on edge devices with resource constraints.
- Simulations involve comparisons between several pre-trained models, including lightweight ones, as well as comparisons with state-of-the-art works and quantization results.

The remainder of the paper is organized as follows. Section II presents the preliminaries while Section III presents the overall workflow. Section IV explains the experimental settings and result discussions. Finally, Section V discusses the conclusion and future works.

II. Preliminaries

A. Post-training Quantization

Post-training quantization refers to the process applied after the model completes its preliminary training, making it more straightforward than quantization-aware training, as it doesn't entail any mutation during training. In contrast, post-training quantization mutates the trained model into a better, tighter form by lowering the weights' precision & activation from thirty-two-bit floating points into a fewer-bit representations like 8-bit or 16-bit. In this work, we used post-training quantization-based dynamic range quantization, which quantized the weights of the model from thirty-two bits (float 32) into an eight-bit integer (INT8), without altering the float32 activations. Such a quantization process can provide a balance between a diminished memory footprint and inference performance, without requiring a calibration dataset.

III. Workflow

Fig. 1 illustrates the overall pipeline of our proposed system, showing how images move through each stage and how the INT8 model is obtained after post-training quantization. Metric values are computed for both the FP32 and INT8 versions. As discussed in Section IV, the most effective pre-trained CNN serves as the backbone, trained in floating-point precision and later converted to its quantized form. The process begins with data preparation, which involves collecting images from an uncontrolled dataset, followed by resizing, normalization, and standardization to ensure consistency. The training data are then split and augmented using flips, rotations, zoom, and contrast adjustments to improve robustness and diversity.

Post-preprocessing, our floating-point model has been subjected to training. The ImageNet-based pre-trained deep CNN model undergoes fine-tuning using actual images and those obtained after augmentation. The fine-tuning process happens in a precision of floating-point thirty-two bits for maintaining good fidelity while doing backpropagation. Throughout the training, monitoring of performance happens via validation data and other hyperparameters listed in Table I.

After training, post-training quantization is applied to make the model efficient for resource-constrained devices. While floating-point models offer higher accuracy, they are computationally expensive. In this process, a small calibration set (drawn from training or validation data) is passed through the model to collect per-layer activation statistics, which are then used to quantize weights and activations to 8-bit integers, thereby reducing memory usage while maintaining accuracy close to the original. Comparisons among quantized and floating-point showcase essential differences. Although post-training quantization can lead to a slight deterioration in performance metrics, our simulation highlights that careful calibration can maintain levels of performance that are approximately near the actual ones. Finally, both post-training quantization and floating-point model performances have been compared on the test set to measure multiple metrics (refer to Section IV-C) in addition to memory footprint. Empirical outcomes also confirm that quantization can not only help retain comparable accuracy but also furnish notable gains in efficiency, which align with our original goal of achieving edge deployment.

Furthermore, aside from post-training quantization (PTQ), we have also experimented with quantization-aware training (QAT) for accuracy preservation while converting to INT8.

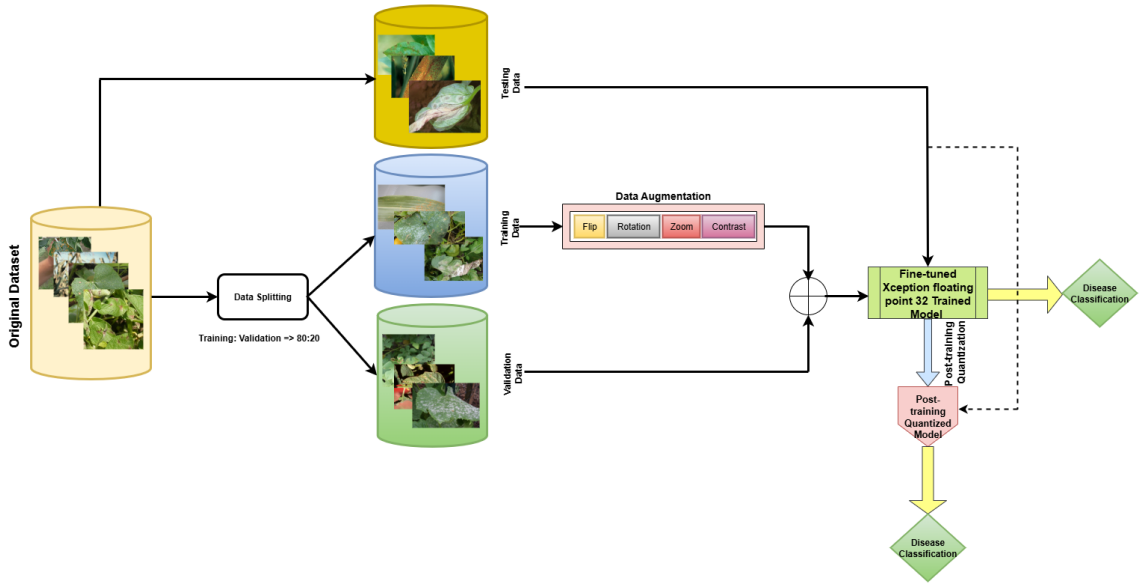

Fig. 1. Overall workflow: plus symbol represents concatenation; Data splitting; Performance metric evaluation; INT8 model after post-quantization training

To boost the transparency factor, we integrate the Gradient-weighted Class Activation Mapping (Grad-CAM) into the proposed pipeline as a means for post-hoc interpretability. After the FP32 and INT8 models reach their acquisition stage, Grad-CAM depictions help focus on specific leaf regions that contribute the most to each prediction class. These explainability highlights instill better confidence and trustworthiness of the proposed solution.

IV. Experiments, Results, And Discussions

A. Dataset Description

The PlantDoc data [15] was conceived for the detection of plant diseases and is built on real-world scenarios. This annotated image dataset comprises 13 crops with 27 disease classes, totaling $2,598$ high-resolution images. Accumulated under natural atmosphere, the data comprises complicated backgrounds, illuminating divergences, and occluded leaves, making it approvingly suggestive of real-world applicability. Because of its various types of diseases and distinctive ecological circumstances, this dataset has found wide usage for evaluating the DL-based models in the detection of plant diseases, making it an invaluable asset for research on agricultural diseases [16]. According to the authors in [17], PlantDoc was collected under non-controlled settings, making it the preferred choice for efficiency checking in this work.

B. Settings

Simulations were conducted to classify disease categories across all 13 crops in the PlantDoc dataset. Since PlantDoc is class-imbalanced [18], class weighting was applied. A pre-trained deep CNN model was fine-tuned on the dataset using TensorFlow with the Keras functional API, while validation loss was continuously monitored. The training hyperparameters are listed in Table I. Using our original pre-trained CNN model, Xception, originally trained on the ImageNet dataset, we performed partial fine-tuning to ensure its smooth adaptability to the PlantDoc target domain. In particular, the layers between the 5th and 41st layers have been unfrozen and retrained, while prior ones have been kept frozen for preserving the generic feature representations. The output feature maps were then followed by a layer applying Global Average Pooling, a dense layer with 512 units, dropout layers obtained using hyperparameter optimization, and finally a layer with softmax activation, where the number of neurons corresponds to the number of disease categories.

We adopted a two-staged training mechanism. In the first stage, Bayesian optimization was employed to explore various hyperparameter combinations across 10 trials, with each trial spanning 5 epochs to facilitate the efficient identification of promising configurations. The hyperparameters under consideration include the learning rate, optimizer type, dropout rate, and batch size. In the subsequent stage, the best-performing configuration from the search was then retrained for 20 epochs, incorporating early stopping, adaptive learning rate reduction, and checkpointing of the model to promote stable convergence and prevent overfitting. Early stopping halted training if validation loss remained unchanged for five consecutive epochs, and the best model was saved for prediction. All experiments were run on a Tesla P100 GPU (16 GB RAM) in Google Colab. To enhance generalization, various data augmentation operations and their values are detailed in Table I. The training data were split 80:20 for validation, with all input images resized to (299×299). After training, the model was evaluated on the test set, and performance metrics were recorded.

C. Performance Metrics

Standard metrics have been utilized for evaluating and comparing our model's performance with that of several

TABLE I
HYPERPARAMETERS AND DATA AUGMENTATION SPECIFICATIONS

Hyperparameters		Data Augmentation	
Name	Value	Name	Value
Batch size	32	Random flip	horizontal
Dropout	0.4	Random rotation	0.1
Epochs	20	Random zoom	0.1
Learning Rate	0.00038	Random Contrast	0.1
Optimizer	adam	-	-
Early stopping	5	-	-

state-of-the-art models. These metrics include accuracy (Acc), precision (Pre), recall (Rec), f1-score (F1), Cohen's kappa score (kappa), and area under the receiver operating curve (AUC).

D. Simulations

TABLE II
PRE-TRAINED CNN MODELS WITH PARAMETER COUNT AND SIZE ON DISK [13]

Model	Top-1 Accuracy	Parameters (in Mil)	Size (in MB)
ResNet50	74.9%	22.9	98
ResNet152	76.6%	60.4	232
InceptionV3	77.9%	23.9	92
DenseNet121	75.0%	8.1	33
DenseNet169	76.2%	14.3	57
Xception	79.0%	22.9	88
NASNetMobile	74.4%	5.3	23
NASNetLarge	82.5%	88.9	343
MobileNet	70.4%	4.3	16
MobileNetV2	71.3%	3.5	14
EfficientNetV2S	83.9%	21.6	88
EfficientNetV2L	85.7%	119	479

1) Selection of Pre-trained Models: Considering the documented choices of parameter count and disk size as among the few paramount factors for edge deployment, they have been given substantial weight when comparing appropriate models in terms of performance metrics. Although accuracy has been a widely accepted measure, we also consider balancing it with parameters and disk size when comparing our work to others. Accordingly, this study selects models from various families, including ResNets, DenseNets, NASNet, MobileNet, and EfficientNets, for comparison. Table II presents the output accuracy, parameters (in millions), and disk size for different CNN models, specifically those in bold.

2) Comparisons with Other Pre-trained Models: First up, we performed comparisons between multiple pre-trained CNN models. The criteria for selecting specific models have been attributed to model complexity and inference time, while maintaining a balance for a comparative analysis [13]. In this context, Table III presents the comparative analysis among multiple models. According to the table, EfficientNetV2S ranks second in terms of accuracy and precision metrics, with a notable distance $\sim$ 10 % from the Xception model. In contrast, the lightest model, MobileNetV2, performs the worst, implying its inability to learn effectively from the challenging PlantDoc dataset. Meanwhile, the most parameter-heavy model, NASNetLarge, comes in distant fourth, suggesting its impairment is much like that of MobileNetV2. Thus, a balanced model while achieving comparable metrics has been the driving force behind this work. Furthermore, we discuss the number of trainable parameters as a key objective behind the utilization of our proposed fine-tuned Xception model, which surpasses the other compared models by margins of at least 5% for all metrics, making it the default choice in this work.

TABLE III
COMPARISONS WITH OTHER PRE-TRAINED CNN MODELS

Model	Acc	Pre	Rec	F1
DenseNet121	51.84%	53.99%	55.51%	53.55%
EfficientNetV2S	52.97%	55.71%	52.97%	52.21%
MobileNetV2	18.22%	18.58%	18.22%	18.39%
NASNetLarge	45.8%	46%	48%	46%
Xception	**58.90%**	**65%**	**59%**	**60%**

TABLE IV
COMPARISONS OF TRAINABLE PARAMETERS

Model	Trainable (in Mil)	Total (in Mil)	Memory (in MB)
DenseNet121	1.34	7.3	27.87
EfficientNetV2S	0.33	20.67	78.84
MobileNetV2	0.33	2.59	9.89
NASNetLarge	0.12	85.02	324.35
Xception	1.06	21.92	83.64

3) Comparison of Trainable Parameters: In this section, we compare the CNN models based on the number of trainable parameters. Additionally, we also highlight the total number of parameters, apart from the model's size on disk, in Table IV. All parameter counts have been reported in millions, with model size in megabytes. From the table, we observe that MobileNetV2 and DenseNet121 incur the least number of total parameters and comparable sizes. Given our aim to deploy on the edge, MobileNetV2 gets the edge, but fails miserably in terms of performance metrics (refer to Table III). Moving on to the next, both EfficientNetV2S and Xception achieve good values, except for NASNetLarge, which has the worst values. As Xception achieves the best values, its model size is quite substantial. Thus, to reach that level, we focused on quantization and studied its performance.

4) Comparisons with State-of-the-art Works: This section compares our proposed fine-tuned model with multiple literary works on the PlantDoc dataset. Accordingly, we present the statistics in Table V, wherein our work achieves the best values for most metrics, except for model size and trainable parameters. The best values for each column have been boldfaced. Although the work by Karthik et al. [19] boasts the least model size, it fails miserably on all other metrics, suggesting its weakness in effective learning. Our approach requires a trainable parameter count that is approximately equal to theirs, but ours exceeds theirs by a significant margin of $\sim 45\%$ across all metrics. While Thakur et al. [23] upped the ante with the least trainable parameter count, our model coins improvements of $\sim$ 40 % over it. While Chen et al. [24] secures the second best, our model overpowers it by

TABLE V
COMPARISONS WITH OTHER STATE-OF-THE-ART WORKS

Model	Acc	Pre	Rec	F1	AUC	Kappa	Trainable Param (in Mil)	Memory(MB)
Karthik et al. [19]	10.17%	20.45%	3.81%	6.42%	58.21%	0.06	0.72	**2.8**
Chen et al. [20]	45.76%	54.14%	41.53%	47%	89.63%	0.46	2.06	16.8
Thai et al. [21]	42.8%	54.79%	33.9%	41.88%	88.43%	0.41	85.79	327.5
Chen et al. [22]	46.19%	49.77%	44.92%	47.22%	84.9%	0.44	3.82	14.8
Thakur et al. [23]	19.49%	19.82%	18.64%	19.21%	63.65%	0.16	**0.49**	27.7
Chen et al. [24]	48.31%	63.35%	43.22%	51.38%	92.3%	0.46	2.06	16.9
Yang et al. [25]	14.83%	50%	10%	16.66%	75.08%	0.11	7	27.4
Proposed	**58.9%**	**65%**	**59%**	**60%**	**96.4%**	**0.57**	1.06	83.64

a majority gain of $\sim$ 10 % while incurring over a million fewer parameters. Interestingly, the plant disease detection metrics across the compared works seem to be coherent with the fact that they struggle to achieve good results on the PlantDoc dataset, highlighting the complexities and challenges involved in modeling the data. Thus, choosing it amplifies the generalization factor to real-world settings, and the values it obtains demonstrate our model's capacity for effective feature discrimination. Finally, with model size on disk being a significant barrier to real-world deployment, we move to quantization in our next section.

5) Quantization Analysis- Post-Training vs. Quantization-Aware Training: First, the Table VI showcases the aggregated outcomes from quantization via PTQ and QAT. For PTQ, the proposed model sustains a limited decline for different metrics, like 1.7%, 4%, and 3%, for accuracy, F1-score, and precision & recall, both, while sustaining a significant drop in model size ($\sim$ 74% from 83.64 MB to 21.55 MB). Interestingly, the corresponding INT8 model post-PTQ also maintains its competitive edge by handsome margins of $\sim 9\%$ and $\sim 5\%$ for accuracy and F1-score, respectively, over second-best state-of-the-art works in Table V. For a fair comparison of the prowess of INT8 quantization results, we also experimented with QAT using similar Bayesian-optimized hyperparameters as those used to achieve the PTQ. For QAT, the INT8 model gives a neck-to-neck fight in terms of accuracy and Cohen's kappa to the FP32 model, while enhancing precision by over 1.7%. In contrast to PTQ, QAT presents a more satisfying quantization path backed by reduced deterioration, making it worthy for reliable low-precision inference. Thus, in general, both PTQ and QAT reduce the model size by a factor of 4, leading to a significant boost in the deployability potential for environments with memory-constrained IoT devices. Additionally, QAT presents enhanced robustness and performance metric stability, paving the way for it to be the preferred choice whenever a minimal computational cost overhead can be accepted during training.

6) Model Interpretability Via Grad-CAM: In this section, we focused on the interpretability and practical adaptability aspects of the proposed model to real-world agricultural settings. Accordingly, we incorporate Gradient-weighted Class Activation Mapping (Grad-CAM) as a suggestive tool to supplement the visualizations that highlight discriminative regions contributing toward the model's predictions. In this context, we present Fig. 2, which showcases the Grad-CAM-based pictorial representations of sample images, including blueberry and corn gray leaf spot from the PlantDoc dataset. In this context, Fig. 2 presents a set of three images, with the left one being the original input, the middle one the Grad-CAM, and the overlay heatmap to the extreme right. The activation heatmaps and their corresponding overlays reveal that our model focuses on biologically relevant features rather than background noisy areas. Furthermore, the actual and estimated class labels have also been provided. With accurate predictions, the pictorial representations demonstrate that our model makes decisions based on influential disease symptomatic areas, thereby enhancing the clarity and reliability for use in real-world settings.

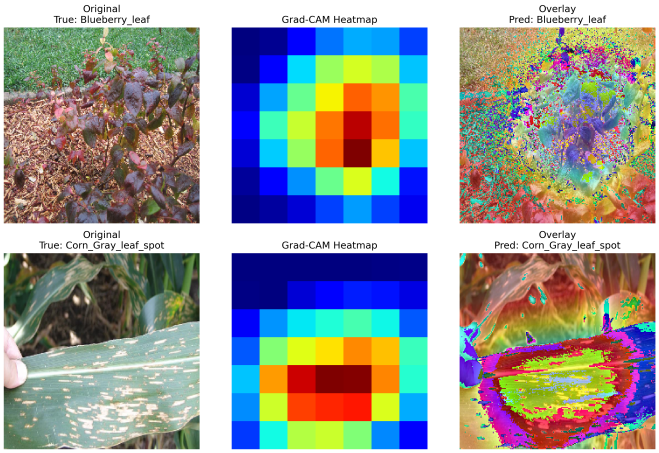

Fig. 2. Grad-CAM based Interpretability analysis of Proposed Model.

V. CONCLUSION AND FUTURE WORKS

This work addresses three crucial considerations: i) validation on real-world field images; ii) good metric values; and iii) fewer parameter count & model size, a rare feat, which must be fulfilled when deploying complex models at the edge to achieve smart agriculture. When dealing with vision-based datasets for plant disease detection, solutions often involve parameter-heavy models with a substantial memory footprint on disk. In response, we proposed a transfer-learning-based fine-tuning approach, enabled by a pre-trained CNN model, Xception, which was validated on a challenging, uncontrolled dataset, PlantDoc. Compared to multiple CNN models, the

TABLE VI
QUANTIZATION RESULTS FOR PTQ AND QAT

(a) Post-Training Quantization (PTQ)							
Model	Acc	Pre	Rec	F1	AUC	Kappa	Memory(MB)
FP32 (Before)	58.9%	65%	59%	60%	96.4%	0.57	83.64
INT8 (After PTQ)	57.2%	62%	56%	56%	95.6%	0.55	21.55
(b) Quantization-Aware Training (QAT)							
Model	Acc	Pre	Rec	F1	AUC	Kappa	Memory(MB)
FP32 (Pseudo-QAT)	55.0%	59.89%	55.24%	53.75%	95.85%	0.5282	83.64
INT8 (QAT)	55.0%	61.80%	55.26%	53.51%	94.91%	0.5282	**21.55**

Xception model obtained substantial gains for all the standard performance metrics to the tune of $\sim 10\%$, which shoots up to $\sim 14\%$ for recall when done with state-of-the-art works. The above statistics highlight the shortcomings of existing literature in failing to adapt to real-world agriculture, a phenomenon that our method addresses. Next, we introduced quantization post-training, reducing the model size by $\sim 74\%$ while maintaining a comparable accuracy (with a slight drop in accuracy of a mere $\sim 1\%$), making it a potential for real-world deployment. Apart from post-training quantization, a quantization-aware training setup and Grad-CAM-based visualization explainability were integrated for strengthening the interpretability and robustness. Although our approach achieves better metric values, there are specific fulfillments that we will address in our potential future work. One such case involves experimenting with cross-dataset generalization, where a model trained in one setting can be adapted to perform well on another dataset with a different condition, both of which belong to the same domain. We will further focus on quantization as a factor that affects the performance metrics in the above scenario.

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